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FOOD QUALITY DESIGN

MSc Thesis

Mapping Gas Permeability of Sustainable Packaging Materials to Link Food Barrier Needs by Clustering Algorithms

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Chair group: Food Quality and Design
Student name: Ting Yu Yeh (1342746)
Supervisor name: Deniz Turan Kunter
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Abstract

Food packaging is critical for ensuring food safety, quality, and shelf life. However, the increasing environmental concerns surrounding conventional packaging have driven interest in sustainable alternatives. A major challenge in adopting sustainable packaging materials is their low barrier properties, which not meet the preservation requirements of certain food products and can lead to increased spoilage and waste. This study aims to address the challenge by employing clustering algorithms to classify sustainable packaging materials based on their gas permeability characteristics, oxygen transmission rate (OTR) and water vapor transmission rate (WVTR). A comprehensive dataset from 49 scientific studies (2000–2016) that spans a wide range of sustainable materials with corresponding OTR and WVTR values was compiled. Clustering techniques including K-Means, Gaussian Mixture Model (GMM), and Density-Based Spatial Clustering of Applications with Noise (DBSCAN) were applied on gas permeability. The results revealed that DBSCAN is the most effective clustering algorithm for this dataset, with highest Silhouette Score (0.900) and lowest Davies-Bouldin Index (0.388). The findings indicated that many sustainable packaging materials exhibit high OTR and WVTR values, making them unsuitable for certain food categories. However, nanocomposite-based materials demonstrate enhanced barrier properties, highlighting their potential for improving the performance of sustainable packaging. This data-driven framework offered a tool for matching of packaging materials to specific food requirements, thereby reducing spoilage and waste while assessing the viability of sustainable alternatives to conventional plastics. Future research should expand the dataset to include additional material properties for a more comprehensive comparison between sustainable and conventional packaging solutions.

Keywords: oxygen transmission rate, water vapor transmission rate, DBSCAN, barrier properties

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Abbreviation

WVTR	Water Vapor Transmission Rate
OTR	Oxygen Transmission Rate
GMM	Gaussian Mixture Model
DBSCAN	Density-Based Spatial Clustering of Applications with Noise

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1. Introduction

1.1 The Role of Permeability in Food Packaging

Food packaging plays a crucial role in ensuring the safety, quality, and shelf life of food products. By acting as a protective barrier against external factors, such as moisture, oxygen, and contaminants, it reduces food waste by preserving the freshness and nutritional value of the food products (Marsh & Bugusu, 2007). To maintain the shelf life of the products, packaging materials must exhibit appropriate gas-barrier properties. The protection effectiveness of these materials depends on their permeability to gases and vapors, which is defined as the transmission of substances through a material and involves four key processes: sorption on the surface, dissolution within the material, diffusion across the material, and desorption from the opposite surface (Choudalakis & Gotsis, 2009; Gajdoš et al., 2000).

In the field of food packaging, the most critical permeability metrics are the water vapor transmission rate (WVTR) and oxygen transmission rate (OTR), as they significantly affect the shelf life of food products (Tyagi et al., 2021). The process of moisture sorption has the potential to induce chemical or physical instability. In addition, the loss of moisture from products through permeation during storage can also lead to changes in their quality (Chen & Li, 2008). For example, lactose crystallization can occur in milk powder without proper packaging due to moisture sorption (Jouppila & Roos, 1994). Oxygen ingress contributes to food spoilage through lipid oxidation in fatty foods, further reducing shelf life, so high-fat foods, such as cheese, require high oxygen barrier packaging. Therefore, it is essential to match the barrier properties of packaging materials to the requirements of food groups based on the food characteristics to minimize the food waste.

1.2 The Emergence of Sustainable Packaging Solutions

Plastics are widely used as food packaging materials today because of their versatility. They can be molded, heat-sealed, and most importantly tailored to provide varying degrees of barrier protection, depending on the polymer composition (Marsh & Bugusu, 2007). In Europe, approximately 40% of all food products are packaged in plastic (Thijs Geijer, 2019). However, approximately 95% of plastic packaging materials, primarily made from petroleum-based polymers such as low-density polyethylene (LDPE), high-density polyethylene (HDPE) and polyethylene terephthalate (PET). They are not recycled and are discarded of after single use, costing the global economy an estimated $80–120 billion annually (The New Plastics Economy, 2017). According to Sander Defruyt (2019), only 2% of plastic packaging is recycled into new packaging material, while the rest is incinerated or disposed in landfills and the environment. Therefore, there is a shift towards sustainable alternatives to reduce waste and minimize carbon footprint.

According to the Web of Science database, publications on sustainable packaging materials have increased significantly over the past 10 years, from 70 to 1,300 (**Figure 1**). These sustainable alternatives are generally classified into three main categories: reusable, recyclable and biodegradable packaging materials(Arvanitoyannis et al., 2014). Reusable packaging refers to materials that can be reused after cleaning, like glass. Recyclable packaging refers to materials that can be reprocessed and reused, such as paper. Biodegradable packaging refers to those materials that can be decomposed in the environment without causing damage, such as cotton bags. Ncube et al. (2020) mentioned that the food packaging industry is now in looking for biodegradable packaging that is lightweight to reduce materials use, waste and also transportation costs to become more sustainable.

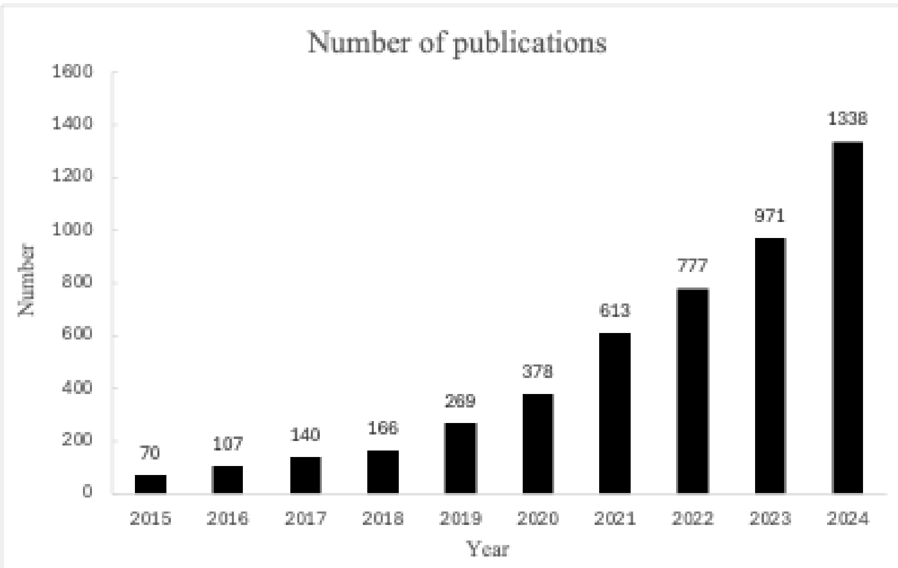

Figure 1. Number of publications under keyword "sustainable packaging material" in Web of Science

1.3 Problems Statement: Challenges in Sustainable Packaging Adoption

In Guillard et al. (2018)'s study, they highlighted the two key challenges in food packaging and sustainability that limit the large-scale market adoption. First, while extensive research has focused on the development of bio-based packaging materials derived from renewable and biodegradable sources, these materials often exhibit inferior barrier and mechanical properties compared to conventional plastics (Ivanković et al., 2017; Sangroniz et al., 2019). Since many biodegradable polymers are hydrophilic, their permeability to gases tends to increase under high humidity due to water absorption (Ashok et al., 2016; Lentschat et al., 2021). These properties indicate that such materials may not provide the same level of protection against moisture and gas permeation as conventional materials. Consequently, maintaining the shelf life and quality of food products becomes challenging, potentially leading to increased food waste, which is contrary to the primary purpose of food packaging. Second, there is a lack of tools to help stakeholders match packaging properties with specific food requirements. While decision support tools, such as those from the Netherlands Institute for Sustainable Packaging (KIDV), are available to predict the environmental impact of packaging based on factors such as circularity, material recyclability, and life cycle assessment (LCA) (Schaarsberg et al., 2024), there is still a need for tools that match gas transfer properties with requirements for different food categories. For example, instant coffee requires very low WVTR and OTR to prevent moisture-induced clogging and oxidation-related flavor loss (Rankin et al., 1958). In contrast, fresh produce, such as fruits and vegetables, require higher OTR and WVTR to allow for post-harvest respiration. High-fat foods such as cheese require low oxygen permeability but can tolerate higher water vapor permeability (Trinh et al., 2023). The absence of such specialized decision tools hinders stakeholders from effectively matching sustainable packaging materials to the unique barrier needs of different food products

While existing studies have identified suitable biodegradable materials for food groups such as biodegradable chitosan-cellulose and polycaprolactone film effective for fresh products like shredded lettuce (Makino & Hirata, 1997) and polyethylene with corn starch for preservation of lean beef (Strantz & Zottola, 1992), these insights are often generalized and lack information on conditions and permeability information. These studies focus primarily on the development of novel materials or the testing of individual materials in isolation, often overlooking a holistic, comparative approach. As a result, many food products remain unsuitable for currently available biodegradable packaging. No comprehensive study has systematically matched sustainable packaging materials to food groups based on gas permeability requirements. Without a systematic understanding of which sustainable materials offer barrier performance comparable to conventional plastics for specific food types, inappropriate packaging choices may compromise food safety, quality, and shelf life. As a result, there is an urgent need for a systematic method to match material gas permeability to specific food requirements.

1.4 Research Aim and Novel Contributions

To address this gap, this study aims to develop clustering algorithms to group sustainable packaging materials according to their gas permeability characteristics namely oxygen and water vapor permeability, then identify which materials are best suited to the barrier needs of different food groups.

Unlike previous studies that have examined materials in isolation, our research innovatively applies data-driven clustering algorithms to systematically classify sustainable packaging materials based on gas permeability.

1.5 Research Questions

Therefore, the research is guided by two key questions:

1. **“Which clustering algorithms most accurately classify sustainable packaging materials based on their gas permeability? “**

2. **“Which sustainable materials are best suited for packaging different food categories based on barrier needs?”**

By answering these questions, the study seeks to provide actionable insights into which materials are most suitable to meet the barrier needs of various food groups based on dataset obtained from 49 scientific studies, published between 2000 and

2016 (Lentschat et al., 2021), with the potential to encourage a transition to more environmentally friendly packaging solutions tailored to specific applications in the food industry.

2. Theoretical Foundation

This section aims to provide the theoretical background necessary to understand the methodology of the study, focusing on packaging barrier properties and clustering algorithms.

2.1 Packaging Barrier Properties

The barrier properties of a material represent its ability to restrict the transmission of vapors, gases, temperature and other external factors from its surroundings (Abdul Khalil et al., 2019). The transmission process of oxygen and water vapor through a material is often quantified in terms of transmission rate (TR) or the permeability coefficient (P).

Transmission rate (TR) measures the amount of permeant passing through a material per unit area and time, typically expressed in g/(m^2·day) (Tyuftin & Kerry, 2023). It does not take into account the influence of material thickness. In contrast, the permeability coefficient (P), measured in units such as barrer (10^{-10} cm^3-cm/(cm^2-s-cmHg)), accounts for the material thickness and the partial pressure gradient (Arrieta, 2016).

The transmission rate (TR) can be defined as:

$$TR = \frac{Q}{At} \tag{1}$$

where Q represents the amount of permeant passing through the material, A denotes the area and t is the time.

Permeability coefficient (P), also referred to permeation rate, can be expressed as:

$$P = S \times D \tag{2}$$

where S is the solubility coefficient (e.g. mol/(m^3·atm) or in other unit expressing amount/ (volume × pressure)), and D is the diffusion coefficient in m^2/s.

The relationship between the oxygen permeability coefficient (OPC) and the oxygen transmission rate (OTR) is expressed as (Siracusa, 2012):

$$OPC = \frac{OTR \times l}{\Delta P} \tag{3}$$

where l stands for film thickness and ΔP is pressure difference across the material.

However, since the barrier properties of packaging materials are often reported in the terms of permeability coefficients (P), the study used the equation (3) to convert permeability coefficients to transmission rates in order to consistent with the gas permeability of food products from Trinh et al. (2023) study. This approach enabled a standardized comparison across materials and studies, supporting the clustering and analysis of sustainable packaging materials based on their barrier performance for different food packaging applications.

Understanding these barrier properties is critical, as they form the basis for our subsequent analysis using clustering algorithms to classify sustainable packaging materials.

2.2 Clustering algorithms

Clustering is an unsupervised learning method that groups similar objects on the basis of their attributes. Clustering techniques can be divided into hierarchical, partition-based, density-based, grid-based and model-based algorithms (Mehta et al., 2020). Given that the dataset is neither high-dimensional nor hierarchical, grid-based algorithm and hierarchical algorithm were excluded from consideration.

The goal of clustering is to maximize the similarity within clusters (intraclass similarity) and minimize the similarity between clusters (interclass similarity). Better clustering results are achieved when objects in the same cluster are more alike and those in different clusters are more distinct.

2.2.1 K-Means Clustering

K-Means is a partition-based clustering algorithm widely known for its efficiency and simplicity. It groups data points into k clusters based on Euclidean distance between points, with k specified as a prerequisite.

The optimal number of clusters can be determined using methods like the elbow method or silhouette score(Nanjundan et al., 2019). The algorithm aims to minimize the sum of squared errors for each cluster (Ikotun et al., 2023) as represented by.

$$\min (X) = \sum_{k=1}^{K} \sum\nolimits_{x_i \in c_k} \left|\left| x_i - \mu_k \right|\right|^2 \tag{4}$$

where x_i is data point in cluster c_k, μ_k is the mean of the cluster and K is the number of clusters.

Although K-Means is computationally efficient, it assumes clusters are spherical and equidistant, which may not fit all datasets, including those with non-centroid distributions.

2.2.2 Gaussian Mixture Model (GMM)

GMM is a model-based clustering method that estimate the parameters of multiple Gaussian distribution in the dataset (Mehta, 2020). These parameters with perquisites k clusters include (Quiñones-Grueiro et al., 2019; Reynolds, 2015):

- Means (μ_k) : the center of the cluster
- Covariance (Σ_k): shape and orientation of the cluster
- Mixing Coefficient (π_k): weight of each Gaussian component (sum = 1)

The parameter of GMM were estimated by using Expectation-Maximization (EM) algorithm, which iteratively refines the model parameters through (Xuan et al., 2001):

- Expectation (E-step): Assigns probabilities for each data point belonging to a cluster
- Maximization (M-step): Updates the parameters to maximize the likelihood

Unlike K-Means, which assumes spherical clusters, GMM handles clusters of varying shapes, sizes, and densities, making it more flexible for complex data distributions (Vila & Schniter, 2013).

2.2.3 Density-Based Spatial Clustering of Applications with Noise (DBSCAN)

DBSCAN is a density-based clustering algorithm that does not require a predefined number of clusters as K-Means and Gaussian Mixed Matrix (Ester et al., 1996b). It identifies clusters of arbitrary shapes and can detect outliers or noise points (Mehta et al., 2020). Key parameters of DBSCAN include:

- Epsilon (ϵ): Maximum radius of a neighborhood
- MinPts: Minimum number of points required to form a cluster

This method is particularly suited for datasets where clusters are not centroid based. Given the non-centroid distribution of the gas permeability data, we hypothesize that DBSCAN will outperform traditional centroid-based methods (such as K-Means) in accurately classifying sustainable packaging materials.

3. Materials and methods

In 3.1 section data description shows the overall review of the data and the software that was used to do further modeling and in 3.2 section the limitations of the dataset were discussed.

3.1. Data Description

3.1.1 General Data

This study utilizes a text-mined dataset compiled from 49 scientific articles published between 2000 and 2016 (Lentschat et al., 2021). The raw data, stored in four comma-separated value files, includes information on 7 variables (**Table 1**). Python 3.8 was used for all subsequent data cleaning, feature engineering, and analysis. Detailed data storage protocols are described in **Appendix A** as data management plan.

After merging and de-duplicating the four datasets, the final dataset contained 1,591 entries, including 1,154 symbolic, 403 quantitative, and 394 addimensional entries. Of these, 22 articles contributed 197 entries with gas permeability data (oxygen, carbon dioxide, and water vapor), while the remaining articles provided primarily symbolic data, such as material types and quantities of impact components in the dataset. However, the remaining 27 articles also contained gas permeability data that was not automatically extracted, so manual extraction was performed for ‘oxygen transmission rate’ (OTR), ‘water vapor transmission rate’ (WVTR), ‘temperature,’ and ‘relative humidity.’ As the research data contains too little carbon dioxide permeability data, the carbon dioxide permeability therefore was removed from this research. As a result, the gas permeability data was expanded from 197 to 295 entries.

Table 1. Variables of raw data, including gas permeability characteristics. The article titled "barrier and surface properties of chitosan-coated greaseproof paper" reports a carbon dioxide permeability of 3400 (cm^3 mm)/m^2·atm·day as a quantity characteristic.

Variable	Explanation	Values
Doc	Article title	Barrier and surface properties of chitosan-coated greaseproof paper
DOI	Digital Object Identifier	https://doi.org/10.1016/j.carbpol.2006.02.005
Target	generic concept represented	Permeability

Type	Ontology concept category, symbolic, quantitative or addimensional	QUANTITY
Original_Value	a list of annotated tokens for symbolic data, two lists of annotated tokens for quantitative data	([‘3400’], [‘cm’,’^’,’3’,’mm’,’/’,’(‘,’m’,’^’,2’, ‘atm’,’day’,’)’])
Attached_Value	a list of annotated tokens to disambiguate a measure unit when necessary for quantitative data. None for symbolic data.	[‘carbon’, ‘dioxide’]
Annotator	annotator id	1

3.1.2 Permeability data

This section provides an overview of the permeability data, including measurement conditions, unit standardization, and data distribution.

Measurement Conditions

Measurement conditions have a significantly effect on gas permeability performance. To assess this impact, the distributions of temperature and relative humidity (RH) were analyzed. For oxygen transmission rate (OTR), the most commonly used test temperature was 23°C, while for water vapor transmission rate (WVTR), it was 25°C. Additionally, 50% RH was the most applied condition in both permeabilities (**Table 2**), aligning with standard measurement practices for permeability (Sánchez-Tamayo et al., 2020). These conditions are consistent with established test methods, such as ASTM D3985 for OTR and ASTM E96 for WVTR.

When measurement conditions were missing, standard conditions were assumed: 23°C and 50% relative humidity (RH) for OTR, and 25°C and 50% RH for WVTR. All subsequent analyses were performed under these standardized conditions during feature selection to avoiding the effect of measurement condition variation.

Table 2. Most commonly reported measurement conditions for oxygen transmission rate (OTR) and water vapor transmission rate (WVTR). The most frequent conditions were 23°C and 50% RH for OTR, and 25°C and 50% RH for WVTR. A significant portion of data lacked measurement condition details.

OTR				WVTR			
Temp (℃)	Quantity	RH (%)	Quantity	Temp (℃)	Quantity	RH (%)	Quantity
23	59	50	46	25	87	50	48
25	26	0	26	20	15	90	14
20	20	57	4	23	9	75	11
missing	8	missing	39	missing	22	missing	37

Units

The reported units for barrier properties were not standardized, provided as either permeability coefficients or transmission rates. Across all barrier properties, 34 unique units were identified, which were standardized during data preprocessing. After preprocessing, $\mathrm{cm^3 cm/m^2\ Pas}$ and $\mathrm{gm/m^2\ Pas}$ were the dominant unit reported permeability coefficient in oxygen and water vapor respectively. For transmission rate. $\mathrm{cm^3/m^2 s}$, $\mathrm{g/m^2 s}$ were the most observed unit in oxygen and water vapor. However, some inconsistencies in reporting were identified. For example, $\mathrm{cm^3 cm/m^2 s}$ was incorrectly reported as a transmission rate. In such cases, the reported value was reinterpreted as a permeability coefficient.

Distribution

The distribution of OTR and WVTR was investigated after confirming measurement conditions and was given in **Figure 2.** OTR exhibited a lower average compared to WVTR but had a wider range, with several extreme outliers. The OTR distribution was strongly right skewed, while WVTR showed a slight left skew. These characteristics highlight the need for standardization before modeling.

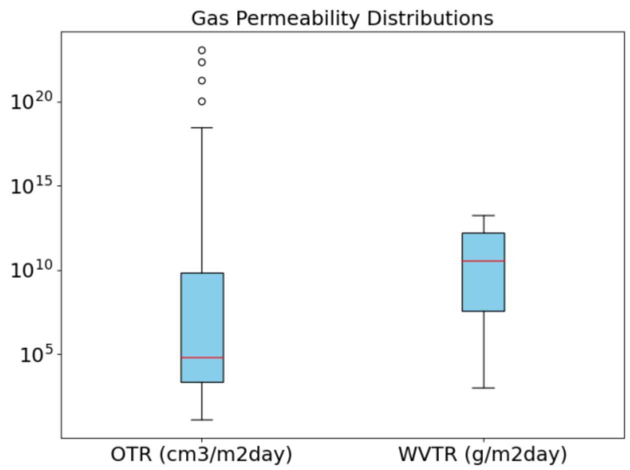

Figure 2. Distribution of oxygen transmission rate (OTR) and water vapor transmission rate (WVTR), both exhibiting skewness, highlighting the need for standardization before modeling

3.2. Limitations of the dataset

The dataset provides some insights for further sustainable materials studies, but there are limitations in the dataset. First, the dataset is limited to 49 scientific articles, which restricts the scope of the research and limits the ability to provide a comprehensive overview of sustainable packaging materials. Additionally, the dataset contains instances where identical values appear in columns, except for the "Original_Value" field. For example, the same “Doc,” “Target,” and “Type” entries linked with different permeability values in “Original_Value”, such as $1.6\ \mathrm{cm^3 \cdot mm/m^2 \cdot 24h \cdot atm}$ and $0.16 \times 10^{-5}\ \mathrm{cm^3 mm/m^2 dayatm}$, making it challenging to accurately interpret the conditions under which these values were recorded. In such cases, further manual investigation into each original article is applied to resolve discrepancies. Moreover, the presence of multiple units for the same type of measurement. For instance, permeability reported in $\mathrm{cm\hat{}3\ mm/m^2 \cdot 24h \cdot atm}$, $\mathrm{cm^3 \cdot mm/m^2 \cdot day \cdot atm}$, and $\mathrm{g/m^2 \cdot s \cdot Pa}$ introduces inconsistencies. Additionally, permeability values are reported under different temperature and humidity conditions, which can further complicate analysis and comparison. As a result, manual intervention or supplementary data processing may be required to address these inconsistencies.

3.3. Research Methods

Before starting modeling, data preprocessing and following feature engineering are necessary to obtain proper features for clustering models. The purpose of data preprocessing includes manage the missing data and unify permeability unit to make the following feature aligned. Feature Engineering consists of text preprocessing, natural language process and lastly feature identification. The purpose of feature engineering is to extract the materials and type of the packaging materials from the article title in variable ‘Doc’.

Clustering algorithms were applied to the logarithmically transformed values of the Water Vapor Transmission Rate (WVTR) and Oxygen Transmission Rate (OTR) to group sustainable packaging materials by their permeability characteristics. Our exploratory analysis revealed that the data distribution is non-spherical, which suggests that methods such as the Gaussian Mixture Model (GMM)—capable of estimating a covariance matrix for each cluster—are more appropriate for capturing complex, elliptical cluster shapes. Additionally, the K-means clustering algorithm will be used as a baseline for performance comparison. Since the optimal number of clusters is unknown, density-based clustering models such as Density-Based Spatial Clustering of Applications with Noise (DBSCAN) will be employed to address this uncertainty (Ester et al., 1996a).

Visualization will be utilized to match the clustering results and gas permeability requirements of food groups. The expected outcome is to identify sustainable materials that can effectively replace conventional plastics in food packaging applications.

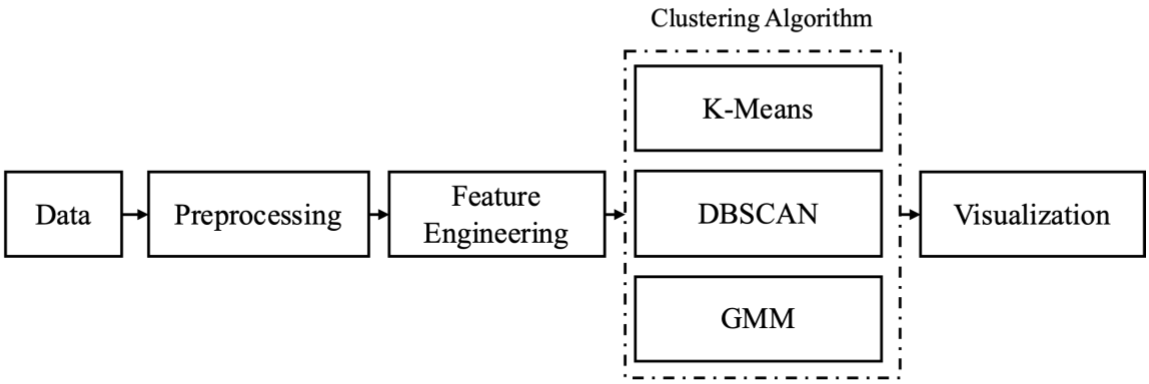

Figure 3. Data processing flowchart

3.3.1. Data preprocessing

Data Cleaning

Although the raw dataset was relatively clean, some duplicate data was identified, and permeability units were not standardized yet. To address this, duplicates were removed, and transmission rate values for oxygen and water vapor were standardized. As detailed in Section 3.1.2, if permeability was reported as a permeability coefficient, it was converted into a transmission rate by multiplying by the material thickness. If thickness was not specified, a default value of 100 μm was assumed. Finally, oxygen transmission rate (OTR) was standardized to $\mathrm{cm^3/m^2day}$ and water vapor transmission rate (WVTR) to $\mathrm{g/m^2day}$.

3.3.2. Feature Engineering

Text Preprocessing

Before performing any analysis, the text data in the 'Doc' column underwent cleaning to remove noise such as punctuation, special characters, and irrelevant symbols. This step ensured that the text is cleaner and easier to process. The cleaned text is then converted to lowercase for consistency.

Natural Language Processing (NLP)

To process the unstructured text data in the "Doc" column, several NLP techniques are applied to identify key components such as sustainable packaging materials, base materials, packaging forms, and modified materials. As the majority of "Doc" entries describe the barrier properties of specific sustainable materials, the analysis begins by refining the text. The following steps are implemented:

1. Preprocessing Text for Parsing: When the word “of” appears in an article title, the text following its first occurrence is extracted. This extracted segment is then analyzed using the SpaCy 3.7.8. library to determine its grammatical structure and syntactic dependencies. This process aids in identifying key components such as base materials and modification type.
2. Dependency Parsing (DP): Dependency Parsing analyzes the grammatical structure of a sentence to determine the relationships between words and their roles in the text (Nivre, 2010). This method was effective for extracting key components in scientific article titles, which typically follow structured grammatical patterns. After parsing, several rules were set to extract key components.

Rules for Extracting Key Components

Base Materials

1. Keyword Matching: if a noun or proper noun matches predefined keywords such as "starch," "chitosan," or "zein," it is identified as the base material.
2. Film/Films: if the word "film" or "films" is present and no base material has been identified, the noun immediately preceding "film" or the direct object (dobj) of "film" is set as the base material. If no preceding noun is available, the function looks for modifiers (e.g., adjectives) to refine the base material, such as in "chitosan films."
3. "Based" Clause: if the word "based" appears in the text, its associated modifier is treated as the base material (e.g., "starch-based").

Material Types: If terms like "nanocomposite" or "coated/coatings" are detected, they are classified as the material type.

Secondary Materials: Keywords such as "montmorillonite," "clay," and "silica" are used to identify secondary materials. If any of these keywords are found, the associated term is extracted as the secondary material.

Feature Identification

After text preprocessing and feature extraction, three key features are extracted from the 'Doc' column, namely based material, modification type, secondary material. For entries where these features could not be automatically extracted using the predefined extraction rules, manual verification was conducted to ensure the correct identification of the relevant features.

3.3.3. Sustainable packaging material clustering

In order to match food groups with sustainable packaging materials, three different clustering algorithms were applied based on their water vapor and oxygen permeability to group materials. The goal is to uncover distinct clusters of materials that share similar permeability properties, which can then be analyzed for suitability in different packaging categories. The data was standardized before entering training step.

Feature Selection

In order to make comparisons with the gas permeability requirements for various food groups identified in previous studies (Trinh et al., 2023), the logarithms of the OTR and WVTR were selected as the primary features for clustering. This logarithmic transformation helps manage the wide range of permeability values observed across different materials, enhancing consistency in the clustering analysis.

Clustering Algorithms

K-Means, Density-Based Spatial Clustering of Applications with Noise (DBSCAN) and Gaussian Mixture Model (GMM) were applied to identify distinct groups of packaging materials based on their gas permeability features.

A pre-defined cluster number k is necessary for K-Means and GMM, and elbow method is widely used to identify clustering number k of K-Means Clustering (Cui, 2020; Syakur et al., 2018). However, the elbow point was ambiguous in the result, so Silhouette Score was applied to determine the optimal clustering number in both K-Means and GMM as shown in **Figure C1** (Shahapure & Nicholas, 2020). As for DBSCAN, according to rule of thumb, the MinPts should larger than dimension+1 and it’s better to be twice as the dimension (Giri et al., 2021). Since the dimension of the data is 2, with only OTR and WVTR, the minimum Points was 4. According to k-distance result

(Naik Gaonkar & Sawant, 2013), epsilon was set as elbow point, which was 1.2 (**Figure C2**).

Optimization

All the temperature and relative humidity conditions are included the primary model. To optimize the algorithm, OTR with standard measurement condition, 23℃ and 50%RH and WVTR with 25℃ and 50%RH were selected for clustering models.

Model Evaluation

In order to respond to the initial research question, which clustering algorithms most accurately classify sustainable packaging materials based on their gas permeability, Silhouette and Davies-Bouldin index and visual domain evaluation are used to determine the performance of the algorithm.

The Silhouette Score is calculated as (Rousseeuw, 1987):

$$s = \frac{b - a}{\max(a,\ b)} \tag{5}$$

where a represents the average distance between a sample and all other points in the same cluster, while b is the average distance between a sample and all other points in the next nearest cluster.

The Davies-Bouldin index is defined as:

$$db = \frac{1}{k}\sum_{i=1}^{k} max\left(\frac{\Delta X_i + \Delta X_j}{\delta(X_i, X_j)}\right) \tag{6}$$

where ΔX_i is the distance within the cluster and $\delta(X_i\ , X_j)$ is the distance between the cluster i and cluster j.

Silhouette Score can measure how well each material fits within its assigned cluster compared to other clusters while Davies-Bouldin Index can assess the similarity between clusters and the separation between them (Davies & Bouldin, 1979).

3.3.4. Visualization

After clustering, the results were visualized using scatter plots that map oxygen and water vapor permeability. These visualizations facilitate direct comparisons between the permeability characteristics of sustainable packaging materials and the specific requirements of eight food groups (e.g., fruits & vegetables, bakery, cheese, etc.), as identified in previous studies (Trinh et al., 2023).

4. Results and Discussion

The primary objective of this research is to develop clustering algorithms for sustainable packaging materials based on their gas permeability properties, oxygen and moisture transmission rates. This can help identify sustainable alternatives to conventional plastics. By grouping materials with similar barrier properties, it will be easier to match them to specific food preservation needs, ensuring food quality and shelf life while reducing reliance on traditional plastics. Therefore, the secondary objective is to identify which sustainable packaging materials are most suitable for 8 different food categories, that is, fresh meat, fruits& vegetables, seafood, bakery products, cheese, peanuts & snacks, coffee and baby food.

4.1 The identified model to classify sustainable packaging materials based on their gas permeability

To deal with missing data, missing data was replaced with the most frequent value which was the condition filtered for standard conditions: 23°C and 50% RH for OTR, and 25°C and 50% RH for WVTR as an optimization procedure. This is done because measurement variations can significantly affect the gas barrier properties of packaging materials (Siracusa, 2012). This filtering ensures the reliability and comparability of the results.

The model performance was evaluated using the Silhouette Score and the Davies-Bouldin Index, with a higher Silhouette Score and a lower Davies-Bouldin Index indicating better clustering performance. DBSCAN exhibited as the best clustering model, showing a clear difference from the other two models (**Table 3**). However, prior to data filtering, the performance between K-Means and DBSCAN shows little difference, highlighting the importance of the quality of the data preprocessing and optimization.

Table 3. Comparison of clustering algorithms (K-Means, DBSCAN, and GMM) based on Silhouette Score and Davies-Bouldin Index before and after optimization. DBSCAN demonstrates superior performance with the highest Silhouette Score and the lowest Davies-Bouldin Index post-optimization, indicating its ability to produce well-defined and compact clusters. The optimization process, involving selecting data under standard conditions, significantly improved the indices for all algorithms, highlighting the effectiveness of parameter tuning.

Metric	Algorithm	Before optimization	After optimization
Silhouette Score	K-Means	0.583	0.680
	DBSCAN	**0.558**	**0.900**
	GMM	0.539	0.705
Davies- Bouldin Index	K-Means	0.673	0.433
	DBSCAN	**0.520**	**0.388**
	GMM	0.772	0.428

To date, no studies have specifically applied clustering algorithms to the gas permeability properties of packaging materials. Existing clustering approaches have predominantly focused on classifying materials based on mechanical or physical properties. For instance, K-Means has been used to categorize plastic polymers based on mechanical properties (Pezer, 2021), while Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) has been utilized to classify mixed-mail items based on attributes such as flexibility and friction, addressing logistical challenges posed by packaging heterogeneity in parcel shipments (Stadlthanner et al., 2024). In the study by Stadlthanner et al. (2024), the adjusted mutual information (AMI) score was used to evaluate the clustering model because the benchmark study was conducted to act as ground truth label. This performance index compares the clustering results with the ground truth labels. In our study, although a similar density-based algorithm was used, the clusters were not determined before the algorithm. Therefore, the silhouette score was chosen as the most appropriate evaluation metric.

In contrast, research on gas permeability has largely focused on predicting the permeability of polymers using machine learning techniques. For example, Wessling et al. (1994) used a neural network to correlate the infrared spectrum of a polymer with its permeability for 33 polymers. More recently, the study of Phan et al. (2024) used a graph neural network trained on both experimental and simulated gas permeability data from 820 polymers to improve prediction accuracy. The limited application of clustering to gas permeability data may be due to the inherent complexity of these datasets, as gas permeability of polymers is influenced by multiple interdependent factors, including polymer structure, free volume, gas-polymer interactions, and processing conditions (De Almeida & Raquel, 2018; Freeman, 1999). The high dimensionality and nonlinear relationships among these parameters pose challenges to effectively structuring the data for clustering analysis.

Despite providing promising insights, the study is constrained by a relatively small dataset, that covers publications from 2000 to 2016. During this period, research on sustainable packaging materials was less extensive, resulting in a narrower range of material properties. Data quality was also a significant challenge, with inconsistencies and missing information for key parameters. The barrier performance of the materials can be influenced by various external and internal factors, such as the sample preparation methods, test conditions as well as micromorphology (Wu et al., 2021). Among these factors, test conditions such as temperature, relative humidity, and material composition significantly affects the reliability and reproducibility of permeability measurements. Additionally, mislabeled data, such as oxygen being mistaken for carbon dioxide, highlighted the need for standardized reporting protocols. Given the proliferation of sustainable packaging studies over the past decade (**Figure 1**), future research can benefit from a broader and more current dataset, provided that issue of data accuracy and consistency issues are addressed.

Since the research used only a small amount of data on gas permeability, expanding the dataset either on gas permeability or beyond gas permeability such as mechanical strength would help to linking packaging properties directly to food product shelf-life. Because the measurement condition was reported ambiguously, standardized and transparent reporting of measurement conditions, material compositions, and processing parameters is essential to ensure reliable, comparable results. Researchers and industry professionals will be better equipped to identify the correct barrier properties of sustainable packaging materials. While this study demonstrates the viability of clustering algorithms in clustering sustainable materials according to gas permeability, future work with more comprehensive or higher quality data is crucial for closing the gap between biodegradable and conventional packaging solutions.

4.2 Matching sustainable packaging materials with barrier requirements of foods

The barrier needs of different food categories, namely, fresh meat, fruits& vegetables, seafood, bakery products, cheese, peanuts & snacks, coffee and baby food were adapted from Trinh et al. (2023). According to the evaluation metrics, the DBSCAN algorithm produced the best clustering outcome with a higher Silhouette Score (0.900) and a lower Davies-Bouldin Index (0.388) for clustering sustainable packaging materials' gas permeability. Therefore, the following matching is based on the DBSCAN clustering model results, which identified distinct groups (blue, orange, green, and purple), while the remaining points were classified as noise. The clustering results of K-Means and GMM can be referred to **Figures B1 and B2**.

As illustrated in **Figure 4,** cluster 1 (blue) partially overlaps with bakery products**,** suggesting that fish gelatin nanocomposites containing montmorillonite could be suitable to use as bakery packaging. Cluster 2 (orange) aligned partly with fruits, vegetables and salads. Although many conventional plastics such as polypropylene appear in cluster 2 (**Table B2**), the presence of chitosan indicates that it may also serve as a sustainable alternative for fresh produce packaging.

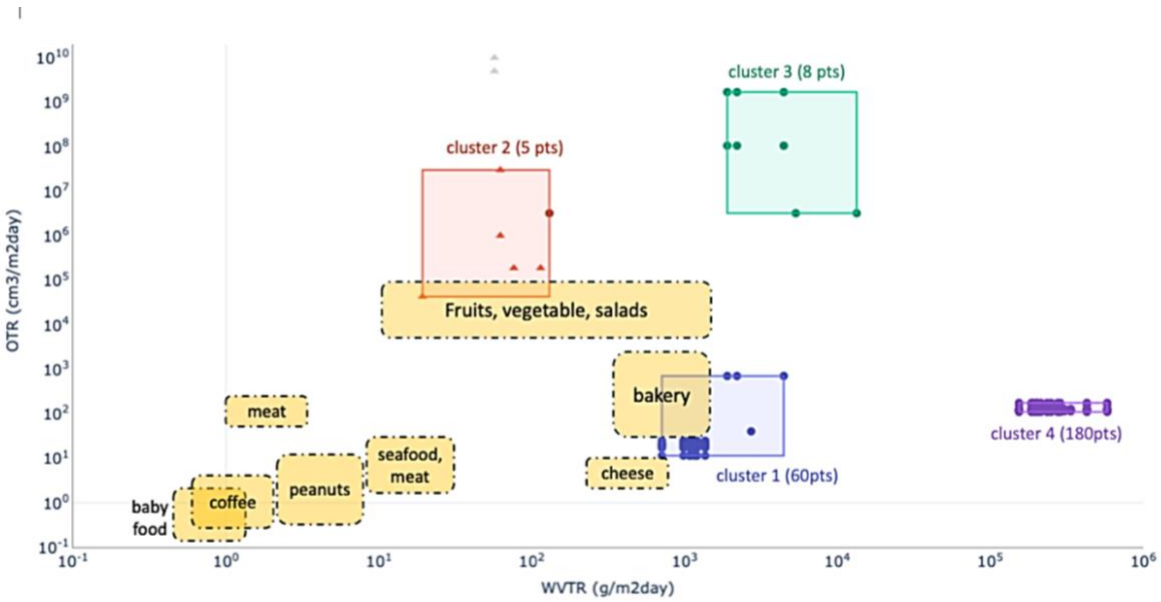

Figure 4. Clusters identified by DBSCAN, color-coded as follows: Cluster 1 (Blue), Cluster 2 (Orange), Cluster 3 (Green), Cluster 4 (Purple). Circular markers (●) represent sustainable materials; triangular markers (▲) represent conventional plastics, whereas grey triangular markers (▲) indicate noise. Because most food groups demand

very low WVTR (water vapor transmission rate) and OTR (oxygen transmission rate), most sustainable materials in this study do not satisfy those requirements, leading to limited clustering overlap. Adapted from Trinh et al.

Meanwhile, other food groups, namely cheese, seafood & meat, peanuts & snacks, fresh meat, coffee, and baby food, show barely overlap with existing sustainable materials (Trinh et al., 2023). These food categories often require low OTR and WVTR, which are generally not met by current sustainable packaging options. This finding reflects a key limitation in sustainable packaging development, as biodegradable polymers often exhibit permeability values one or more orders of magnitude higher than conventional synthetic polymers (Auras et al., 2005; Siracusa et al., 2008). The limited overlap between sustainable materials and food-specific gas permeability requirements highlights a gap in current packaging solutions.

Recent research has explored various barrier enhancement strategies to bridge the gap between sustainable materials and food packaging requirements while addressing environmental concerns. One approach involves coating techniques, such as PLA-coated paperboard with an AlOx layer, which has been shown to reduce both OTR and WVTR (Peelman et al., 2013). Similarly, coating soy protein isolate (SPI) films with PLA showed a reduction in WVTR (Rhim et al., 2007). Another promising strategy is the incorporation of nanoparticles into polymer matrices. Nanocomposites have been widely recognized for their ability to improve the barrier performance of bio-based films. In this study, Cluster 1, which includes nanocomposites, exhibited the lowest OTR among all clusters (**Table 4**), consistent with Trinh et al. (2023) findings. The study of Arora & Padua (2010) showed that PLA combined with montmorillonite-layered silicate significantly improves barrier properties, making it a viable option for food packaging. Additionally, incorporating organically modified montmorillonite (OMMT, Cloisite) into thermoplastic starch (TPS) enhances tensile strength and reduces WVTR (Park et al., 2002).

Table 4. Summary of DBSCAN cluster outcomes, showing the average (± standard deviation) OTR and WVTR for each cluster, along with their dominant base material, type, and secondary material

Cluster	1 (n = 60)	2 (n = 6)	3 (n = 8)	4 (n = 180)
OTR (cm^3/m^2day)	22.74 ± 2.30	733017.02±10.60	126198850±13.46	126.35±1.13
WVTR (g/m^2day)	1146.83±1.35	66.27±1.97	3548.47±1.98	248416.31±1.40
Dominant Base Material	fish gelatin	LDPE	polylactic acid	carrot puree
Dominant Type	nanocomposite	individual	individual	blend
Dominant Secondary Material	montmorillonite	None	None	CMC

Nanocomposites have demonstrated superior barrier properties compared to virgin bio-based materials, with nanofilled materials consistently outperforming their unmodified counterparts (Arora & Padua, 2010). This technology could also be extended to active packaging applications, such as antimicrobial packaging designed to extend the shelf life of food products (Brandelli, 2024). Further research should explore the integration of nanocomposites with sustainable materials as a viable strategy to improve packaging performance. However, before commercial implementation, the potential migration of nanomaterials into food must be thoroughly evaluated to ensure safety and regulatory compliance (Peelman et al., 2013).

5. Conclusion

This study revealed that the density-based clustering algorithm, DBSCAN, is the most effective algorithm for grouping sustainable packaging materials based on gas permeability, with highest the Silhouette Score (0.900) and lowest Davies-Bouldin Index (0.388). Additionally, the results indicate that most sustainable packaging materials have high gas permeability, whereas many food applications require high gas barrier properties. This aligns with previous findings on biodegradable polymers. However, the incorporation of nanoparticles or nanofillers improves barrier properties, where nanocomposite materials outperform single polymers and blends.

5.1 Research Limitation

In this study, several limitations exist. The dataset, derived from publications between 2000 and 2016, was relatively small, restricting the diversity of material properties. Inconsistencies and missing data in key parameters such as temperature, relative humidity, and material composition further impacted the reliability of permeability measurements. Hence, standardized reporting of measurement conditions, material compositions, and processing parameters is essential for improving data reliability and applicability.

5.2 Research Outlook and Recommendation

The study shows that clustering algorithms can effectively classify sustainable materials. Future research could expand datasets and improve data quality to connect material characteristics with food preservation needs better. Incorporating a broader range of sustainable materials, including newly developed biopolymers and advanced coatings, will provide a more comprehensive understanding of their barrier properties. Additionally, extending the dataset beyond 2016 to include recent advancements in material science will improve the relevance of the findings. Including more gas permeability data or additional material properties like mechanical strength and biodegradability or recyclability will enable a more balanced assessment of sustainable packaging performance. Moreover, standardized reporting of measurement conditions, material compositions, and processing parameters are also essential for obtaining reliable and comparable results.

Nanocomposites have improved barrier properties and could be designed with sustainable materials for better performance. The use of active packaging technologies, such as antimicrobial coatings, may also enhance the functionality of sustainable packaging.

Combining different sustainable materials or developing multilayered structures could enhance barrier performance while maintaining environmental benefits. Future studies should explore how hybrid packaging approaches can be optimized for food preservation. Leveraging more advanced machine learning techniques, such as deep learning or reinforcement learning, could improve material classification and prediction accuracy. Additionally, integrating real-world performance data from packaging trials could enhance the robustness of the model. To fully assess the practical impact of sustainable packaging, future research should consider existing and emerging regulatory frameworks. This will help determine the trade-offs between sustainability and performance.

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Data availability

Raw data source from Lenschet study can access via CIRAD open access portal
https://dataverse.cirad.fr/dataset.xhtml?persistentId=doi:10.18167/DVN1/U7HK8J

Code availability

Processed dataset package is available and code can access via GitHub repository
https://github.com/whps0620/Food-Pack-Mapper/tree/main

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Appendix A: Data Management Plan

Heading	Guidance
Organizational context	Thesis: Identifying Sustainable Packaging Solutions for Food Groups Through Clustering of Gas Permeability Chair Group: Food Quality & Design Student Name: Ting Yu Yeh (MSc) Supervisor Name: Deniz Turan Kunter
Short description of your research	This thesis aims to explore the relationship between food types, their specific packaging needs, and the potential of various sustainable materials in meeting these needs. By leveraging a text-mined dataset derived from 50 scientific articles on food packaging composition and gas permeability, this research will focus on clustering food types based on their packaging requirements and identifying promising sustainable materials suitable for different food categories.
Data management rules	The MSc student has control over the data during the research, with the supervisor overseeing and giving feedback on the entire process. Once the student completes the thesis and lefts, the responsibility for data management will be transferred to the chair group, ensuring proper long-term stewardship and adherence to institutional guidelines.
What type of research data will be produced?	The research will generate processed data, models, and detailed documentation of the research process. The dataset will be derived from text-mined data collected from 50 scientific articles on food packaging composition and gas permeability. As part of the research, a clustering algorithm will be developed for data analysis, which will also be a key output. The entire analysis, including data processing and methodology, will be thoroughly documented in Jupyter notebooks.

Software choices (*)	Python 3.8, the widely used software in data science, will be used to clean, analyze and visualize data.
What is the amount of the data, and how will the amount increase in time.	goldenTRANSMAT-all.csv (text/csv) 0.38 megabytes. goldenTRANSMAT-annotator1.csv (text/csv) 0.36 megabytes. goldenTRANSMAT-annotator2.csv (text/csv) 0.03 megabytes. goldenTRANSMAT-annotator3.csv (text/csv) 0.02 megabytes.
Sharing and ownership (*)	The research has generated open data on sustainable packaging materials and their gas permeability characteristics, which could be valuable to researchers, companies, and institutions in food science, packaging technology, and sustainability. I am open to sharing this data, particularly after the research is completed, as it may contribute to advancing sustainable packaging solutions. However, proper attribution and citation will be required when re-using the data. There are no specific funder requirements or embargoes governing the sharing of this data. While no formal agreements with external parties are in place at this time, any future collaboration will involve clear agreements on data usage and sharing to ensure appropriate data management and protection of intellectual property. There are no significant privacy or security concerns with this dataset, as it comprises scientific data related to packaging materials and does not contain personal or sensitive information.

Documentation and metadata (*)	During research **1.** File Naming Conventions: All files, including datasets and notebooks, will follow a clear and consistent naming convention (e.g. cleaned_data_v1.csv and EDA.ipynb). **2.** Metadata and Descriptions: Metadata, descriptions of the dataset, variables, units of measurement, and analysis methods will be thoroughly documented within the Jupyter notebooks. Each notebook will include comments, markdown cells, and code annotations to provide comprehensive explanations for future reference. **3.** Version Control: Changes to both data files and Jupyter notebooks will be tracked using version control (e.g., Git), maintaining a log of all updates and modifications throughout the research. **4.** Storage and Backup: Data will be stored within the Microsoft Teams environment of Wageningen University & Research, For Long-Term Storage • Final Dataset and Notebooks: A final version of the dataset and the accompanying Jupyter notebooks will be prepared. These will include the cleaned data, documented analysis, and any visualizations or outputs. • Archival Format: The data will be saved in non-proprietary formats like CSV, while Jupyter notebooks will be stored as .ipynb files.

	• Secure Storage: The final data and notebooks will be archived in teams. For Data Sharing • Data and Notebook Accessibility: The dataset and Jupyter notebooks will be shared through open repositories or GitHub. Access can be open or restricted, depending on the nature of the data. • User Guide: A user guide will be embedded within the Jupyter notebooks using markdown cells, detailing the dataset's structure, variables, methods, and any limitations. Instructions on how to replicate analyses and use the data will also be included.
Short term storage (*)	Data will be stored in non-proprietary formats for compatibility: • Raw and Processed Data: CSV (.csv) • Documentation and Analysis: Jupyter notebooks (.ipynb), containing embedded metadata and analysis descriptions.

	Data and notebooks will be securely stored in Microsoft Teams with automatic daily backups, along with local backups on the laptop A structured directory will reflect the research stages, specifically for sustainable packaging materials (SPM), including folders for “Raw Data”, “Processed Data”, “Analysis” and “Results”. File naming will follow a consistent format (e.g., SPM_clean_data_version.csv, SPM_EDA.ipynb) for easy retrieval Version control will be managed using Git to track changes in datasets and notebooks, ensuring an accurate modification history. All documentation will be maintained in Jupyter notebooks, featuring markdown cells to explain file purposes and processing steps. A final metadata document will summarize the project and its files for long-term storage.
Long term storage	I plan to store the data in GitWUR for the long term after the conclusion of my research. This is aligned with common practice in Wageningen University and Research, because the data may be of interest for future re-use by other researchers or industry professionals.

Wageningen Graduate School format for a Data Management Plan for research projects and guidance

Appendix B: Clustering Results

Appendix B presents a comprehensive summary of the clustering analysis conducted in this study. It includes detailed results for the DBSCAN model (**Table B1** to **Table B4**) and the outcomes from the two alternative models, K-Means and Gaussian Mixture Model (GMM) are shown as **Figure B1** and **Figure B2**. The appendix provides visualizations of the clusters.

Table B1. The data points in cluster 1

Base Material	Type	Secondary Material	OTR	WVTR
fish gelatin	individual	None	40.28	2731.398
fish gelatin	nanocomposite	montmorillonite	11.44	709.113
fish gelatin	nanocomposite	montmorillonite	11.44	980.502
fish gelatin	nanocomposite	montmorillonite	11.44	1199.364
fish gelatin	nanocomposite	montmorillonite	11.44	1365.699
fish gelatin	nanocomposite	montmorillonite	11.44	1085.556
fish gelatin	nanocomposite	montmorillonite	11.44	1076.801
fish gelatin	nanocomposite	montmorillonite	11.44	1146.837
fish gelatin	nanocomposite	montmorillonite	11.44	1208.118
fish gelatin	nanocomposite	montmorillonite	19.64	709.113
fish gelatin	nanocomposite	montmorillonite	19.64	980.502
fish gelatin	nanocomposite	montmorillonite	19.64	1199.364
fish gelatin	nanocomposite	montmorillonite	19.64	1365.699
fish gelatin	nanocomposite	montmorillonite	19.64	1085.556
fish gelatin	nanocomposite	montmorillonite	19.64	1076.801
fish gelatin	nanocomposite	montmorillonite	19.64	1146.837
fish gelatin	nanocomposite	montmorillonite	19.64	1208.118
fish gelatin	nanocomposite	montmorillonite	17.13	709.113
fish gelatin	nanocomposite	montmorillonite	17.13	980.502
fish gelatin	nanocomposite	montmorillonite	17.13	1199.364
fish gelatin	nanocomposite	montmorillonite	17.13	1365.699
fish gelatin	nanocomposite	montmorillonite	17.13	1085.556
fish gelatin	nanocomposite	montmorillonite	17.13	1076.801
fish gelatin	nanocomposite	montmorillonite	17.13	1146.837
fish gelatin	nanocomposite	montmorillonite	17.13	1208.118
fish gelatin	nanocomposite	montmorillonite	21.42	709.113
fish gelatin	nanocomposite	montmorillonite	21.42	980.502
fish gelatin	nanocomposite	montmorillonite	21.42	1199.364
fish gelatin	nanocomposite	montmorillonite	21.42	1365.699
fish gelatin	nanocomposite	montmorillonite	21.42	1085.556
fish gelatin	nanocomposite	montmorillonite	21.42	1076.801
fish gelatin	nanocomposite	montmorillonite	21.42	1146.837
fish gelatin	nanocomposite	montmorillonite	21.42	1208.118
fish gelatin	nanocomposite	montmorillonite	18.99	709.113
fish gelatin	nanocomposite	montmorillonite	18.99	980.502
fish gelatin	nanocomposite	montmorillonite	18.99	1199.364
fish gelatin	nanocomposite	montmorillonite	18.99	1365.699
fish gelatin	nanocomposite	montmorillonite	18.99	1085.556
fish gelatin	nanocomposite	montmorillonite	18.99	1076.801
fish gelatin	nanocomposite	montmorillonite	18.99	1146.837
fish gelatin	nanocomposite	montmorillonite	18.99	1208.118

fish gelatin	nanocomposite	montmorillonite	24.8	709.113
fish gelatin	nanocomposite	montmorillonite	24.8	980.502
fish gelatin	nanocomposite	montmorillonite	24.8	1199.364
fish gelatin	nanocomposite	montmorillonite	24.8	1365.699
fish gelatin	nanocomposite	montmorillonite	24.8	1085.556
fish gelatin	nanocomposite	montmorillonite	24.8	1076.801
fish gelatin	nanocomposite	montmorillonite	24.8	1146.837
fish gelatin	nanocomposite	montmorillonite	24.8	1208.118
fish gelatin	nanocomposite	montmorillonite	20.73	709.113
fish gelatin	nanocomposite	montmorillonite	20.73	980.502
fish gelatin	nanocomposite	montmorillonite	20.73	1199.364
fish gelatin	nanocomposite	montmorillonite	20.73	1365.699
fish gelatin	nanocomposite	montmorillonite	20.73	1085.556
fish gelatin	nanocomposite	montmorillonite	20.73	1076.801
fish gelatin	nanocomposite	montmorillonite	20.73	1146.837
fish gelatin	nanocomposite	montmorillonite	20.73	1208.118
polylactic acid	individual	None	711.302	2203.925
polylactic acid	individual	None	711.302	1904.099
polylactic acid	individual	None	711.302	4463.444

Table B2. The data points in cluster 2

Base Material	Type	Secondary Material	OTR	WVTR
LDPE	individual	None	189481.965	114.438
LDPE	individual	None	189481.965	76.164
HDPE	individual	None	43264.64	19.26
chitosan	individual	None	3242400.0	130.617
polyvinyl chloride	individual	None	1013243.515	62.157
polyvinyl chloride	individual	None	30397305.456	62.157

Table B3. The data points in cluster 3

Base Material	Type	Secondary Material	OTR	WVTR
polylactic acid	individual	None	106815955.213	2203.925
polylactic acid	individual	None	106815955.213	1904.099
polylactic acid	individual	None	106815955.213	4463.444
polylactic acid	individual	None	1712381540.688	2203.925
polylactic acid	individual	None	1712381540.688	1904.099
polylactic acid	individual	None	1712381540.688	4463.444
chitosan	individual	None	3242400.0	5348.987
chitosan	individual	None	3242400.0	13395.165

Table B4. The data points in cluster 4

Base Material	Type	Secondary Material	OTR	WVTR
carrot puree	blend	CMC	177.453	580509.569
carrot puree	blend	CMC	177.453	427743.893
carrot puree	blend	CMC	177.453	248977.411
carrot puree	blend	CMC	177.453	202053.398
carrot puree	blend	CMC	177.453	185770.066
carrot puree	blend	CMC	177.453	154166.393
carrot puree	blend	CMC	177.453	286446.586
carrot puree	blend	CMC	177.453	276028.754
carrot puree	blend	CMC	177.453	235407.967
carrot puree	blend	CMC	177.453	213784.402

carrot puree	blend	CMC	177.453	200127.413
carrot puree	blend	CMC	177.453	190935.209
carrot puree	blend	CMC	158.281	580509.569
carrot puree	blend	CMC	158.281	427743.893
carrot puree	blend	CMC	158.281	248977.411
carrot puree	blend	CMC	158.281	202053.398
carrot puree	blend	CMC	158.281	185770.066
carrot puree	blend	CMC	158.281	154166.393
carrot puree	blend	CMC	158.281	286446.586
carrot puree	blend	CMC	158.281	276028.754
carrot puree	blend	CMC	158.281	235407.967
carrot puree	blend	CMC	158.281	213784.402
carrot puree	blend	CMC	158.281	200127.413
carrot puree	blend	CMC	158.281	190935.209
carrot puree	blend	CMC	130.092	580509.569
carrot puree	blend	CMC	130.092	427743.893
carrot puree	blend	CMC	130.092	248977.411
carrot puree	blend	CMC	130.092	202053.398
carrot puree	blend	CMC	130.092	185770.066
carrot puree	blend	CMC	130.092	154166.393
carrot puree	blend	CMC	130.092	286446.586
carrot puree	blend	CMC	130.092	276028.754
carrot puree	blend	CMC	130.092	235407.967
carrot puree	blend	CMC	130.092	213784.402
carrot puree	blend	CMC	130.092	200127.413
carrot puree	blend	CMC	130.092	190935.209
carrot puree	blend	CMC	120.374	580509.569
carrot puree	blend	CMC	120.374	427743.893
carrot puree	blend	CMC	120.374	248977.411
carrot puree	blend	CMC	120.374	202053.398
carrot puree	blend	CMC	120.374	185770.066
carrot puree	blend	CMC	120.374	154166.393
carrot puree	blend	CMC	120.374	286446.586
carrot puree	blend	CMC	120.374	276028.754
carrot puree	blend	CMC	120.374	235407.967
carrot puree	blend	CMC	120.374	213784.402
carrot puree	blend	CMC	120.374	200127.413
carrot puree	blend	CMC	120.374	190935.209
carrot puree	blend	CMC	114.421	580509.569
carrot puree	blend	CMC	114.421	427743.893
carrot puree	blend	CMC	114.421	248977.411
carrot puree	blend	CMC	114.421	202053.398
carrot puree	blend	CMC	114.421	185770.066
carrot puree	blend	CMC	114.421	154166.393
carrot puree	blend	CMC	114.421	286446.586
carrot puree	blend	CMC	114.421	276028.754
carrot puree	blend	CMC	114.421	235407.967
carrot puree	blend	CMC	114.421	213784.402
carrot puree	blend	CMC	114.421	200127.413

carrot puree	blend	CMC	114.421	190935.209
carrot puree	blend	CMC	110.044	580509.569
carrot puree	blend	CMC	110.044	427743.893
carrot puree	blend	CMC	110.044	248977.411
carrot puree	blend	CMC	110.044	202053.398
carrot puree	blend	CMC	110.044	185770.066
carrot puree	blend	CMC	110.044	154166.393
carrot puree	blend	CMC	110.044	286446.586
carrot puree	blend	CMC	110.044	276028.754
carrot puree	blend	CMC	110.044	235407.967
carrot puree	blend	CMC	110.044	213784.402
carrot puree	blend	CMC	110.044	200127.413
carrot puree	blend	CMC	110.044	190935.209
carrot puree	blend	CMC	118.973	580509.569
carrot puree	blend	CMC	118.973	427743.893
carrot puree	blend	CMC	118.973	248977.411
carrot puree	blend	CMC	118.973	202053.398
carrot puree	blend	CMC	118.973	185770.066
carrot puree	blend	CMC	118.973	154166.393
carrot puree	blend	CMC	118.973	286446.586
carrot puree	blend	CMC	118.973	276028.754
carrot puree	blend	CMC	118.973	235407.967
carrot puree	blend	CMC	118.973	213784.402
carrot puree	blend	CMC	118.973	200127.413
carrot puree	blend	CMC	118.973	190935.209
carrot puree	blend	CMC	120.199	580509.569
carrot puree	blend	CMC	120.199	427743.893
carrot puree	blend	CMC	120.199	248977.411
carrot puree	blend	CMC	120.199	202053.398
carrot puree	blend	CMC	120.199	185770.066
carrot puree	blend	CMC	120.199	154166.393
carrot puree	blend	CMC	120.199	286446.586
carrot puree	blend	CMC	120.199	276028.754
carrot puree	blend	CMC	120.199	235407.967
carrot puree	blend	CMC	120.199	213784.402
carrot puree	blend	CMC	120.199	200127.413
carrot puree	blend	CMC	120.199	190935.209
carrot puree	blend	CMC	120.462	580509.569
carrot puree	blend	CMC	120.462	427743.893
carrot puree	blend	CMC	120.462	248977.411
carrot puree	blend	CMC	120.462	202053.398
carrot puree	blend	CMC	120.462	185770.066
carrot puree	blend	CMC	120.462	154166.393
carrot puree	blend	CMC	120.462	286446.586
carrot puree	blend	CMC	120.462	276028.754
carrot puree	blend	CMC	120.462	235407.967
carrot puree	blend	CMC	120.462	213784.402
carrot puree	blend	CMC	120.462	200127.413
carrot puree	blend	CMC	120.462	190935.209

carrot puree	blend	CMC	122.388	580509.569
carrot puree	blend	CMC	122.388	427743.893
carrot puree	blend	CMC	122.388	248977.411
carrot puree	blend	CMC	122.388	202053.398
carrot puree	blend	CMC	122.388	185770.066
carrot puree	blend	CMC	122.388	154166.393
carrot puree	blend	CMC	122.388	286446.586
carrot puree	blend	CMC	122.388	276028.754
carrot puree	blend	CMC	122.388	235407.967
carrot puree	blend	CMC	122.388	213784.402
carrot puree	blend	CMC	122.388	200127.413
carrot puree	blend	CMC	122.388	190935.209
carrot puree	blend	CMC	124.751	580509.569
carrot puree	blend	CMC	124.751	427743.893
carrot puree	blend	CMC	124.751	248977.411
carrot puree	blend	CMC	124.751	202053.398
carrot puree	blend	CMC	124.751	185770.066
carrot puree	blend	CMC	124.751	154166.393
carrot puree	blend	CMC	124.751	286446.586
carrot puree	blend	CMC	124.751	276028.754
carrot puree	blend	CMC	124.751	235407.967
carrot puree	blend	CMC	124.751	213784.402
carrot puree	blend	CMC	124.751	200127.413
carrot puree	blend	CMC	124.751	190935.209
carrot puree	blend	CMC	127.203	580509.569
carrot puree	blend	CMC	127.203	427743.893
carrot puree	blend	CMC	127.203	248977.411
carrot puree	blend	CMC	127.203	202053.398
carrot puree	blend	CMC	127.203	185770.066
carrot puree	blend	CMC	127.203	154166.393
carrot puree	blend	CMC	127.203	286446.586
carrot puree	blend	CMC	127.203	276028.754
carrot puree	blend	CMC	127.203	235407.967
carrot puree	blend	CMC	127.203	213784.402
carrot puree	blend	CMC	127.203	200127.413
carrot puree	blend	CMC	127.203	190935.209
carrot puree	blend	gelatin	120.899	251428.666
carrot puree	blend	gelatin	120.899	211333.147
carrot puree	blend	gelatin	120.899	188746.589
carrot puree	blend	gelatin	120.899	237333.953
carrot puree	blend	gelatin	120.899	303692.911
carrot puree	blend	gelatin	120.899	335121.494
carrot puree	blend	gelatin	113.546	251428.666
carrot puree	blend	gelatin	113.546	211333.147
carrot puree	blend	gelatin	113.546	188746.589
carrot puree	blend	gelatin	113.546	237333.953
carrot puree	blend	gelatin	113.546	303692.911
carrot puree	blend	gelatin	113.546	335121.494
carrot puree	blend	gelatin	122.300	251428.666

carrot puree	blend	gelatin	122.300	211333.147
carrot puree	blend	gelatin	122.300	188746.589
carrot puree	blend	gelatin	122.300	237333.953
carrot puree	blend	gelatin	122.300	303692.911
carrot puree	blend	gelatin	122.300	335121.494
carrot puree	blend	gelatin	117.135	251428.666
carrot puree	blend	gelatin	117.135	211333.147
carrot puree	blend	gelatin	117.135	188746.589
carrot puree	blend	gelatin	117.135	237333.953
carrot puree	blend	gelatin	117.135	303692.911
carrot puree	blend	gelatin	117.135	335121.494
carrot puree	blend	gelatin	129.566	251428.666
carrot puree	blend	gelatin	129.566	211333.147
carrot puree	blend	gelatin	129.566	188746.589
carrot puree	blend	gelatin	129.566	237333.953
carrot puree	blend	gelatin	129.566	303692.911
carrot puree	blend	gelatin	129.566	335121.494
carrot puree	blend	gelatin	127.553	251428.666
carrot puree	blend	gelatin	127.553	211333.147
carrot puree	blend	gelatin	127.553	188746.589
carrot puree	blend	gelatin	127.553	237333.953
carrot puree	blend	gelatin	127.553	303692.911
carrot puree	blend	gelatin	127.553	335121.494

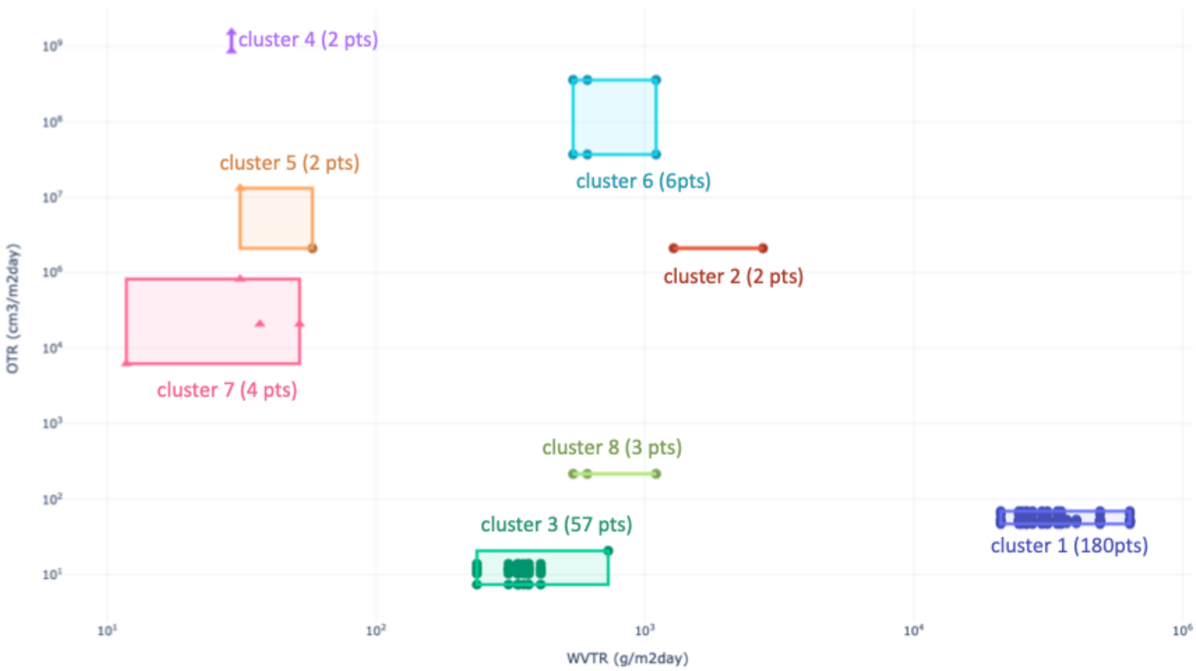

Figure B1. GMM model clustering outcome with 8 distinct clusters

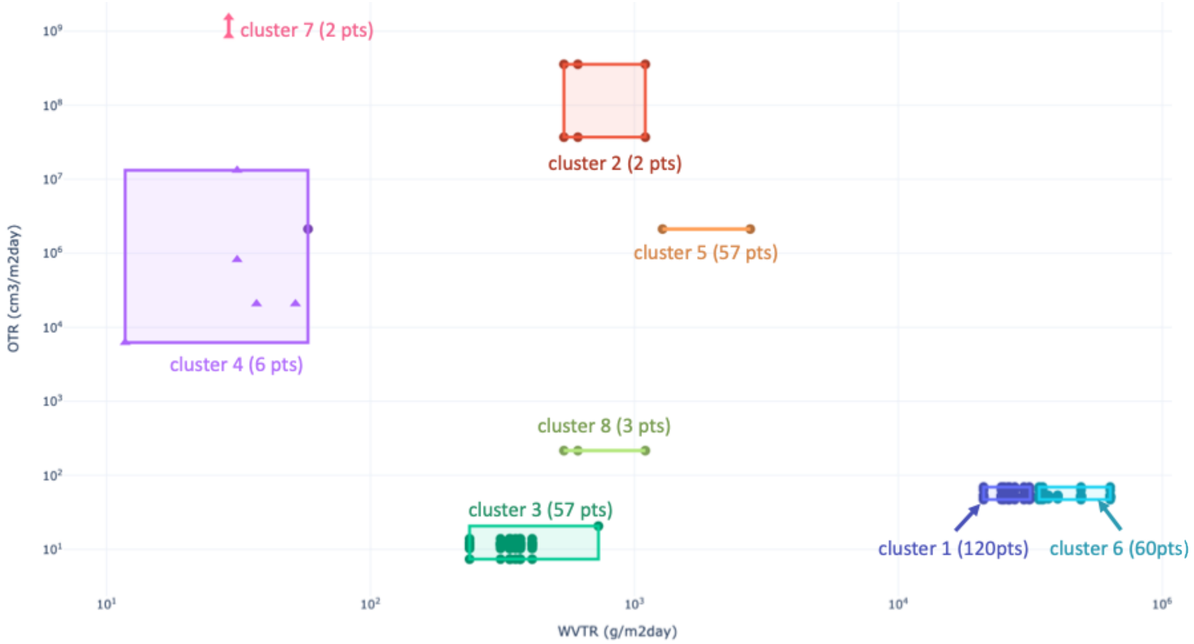

Figure B2. K-Means model clustering outcome with 8 distinct clusters,

Appendix C: Model Tuning Hyperparameter

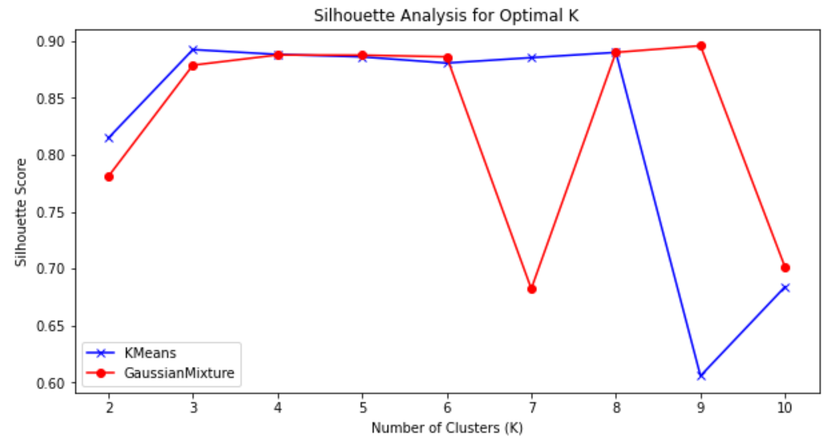

Figure C1. Silhouette Score Analysis for Optimal Cluster Determination in K-Means and Gaussian Mixture Models. Both algorithms achieve their highest silhouette score when the number of clusters is set to 8, indicating that 8 clusters is the optimal configuration for each model.

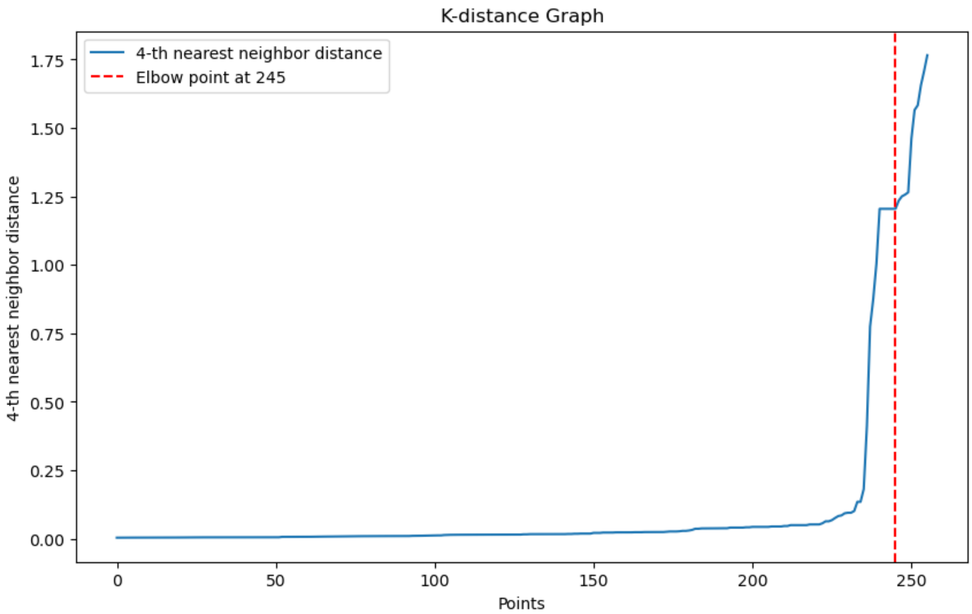

Figure C2. K-Distance Plot for Determining DBSCAN ε Parameter. The “elbow” observed at the 245th point corresponds to an ε value of 1.2, which was selected as the optimal neighborhood radius for the DBSCAN clustering algorithm.